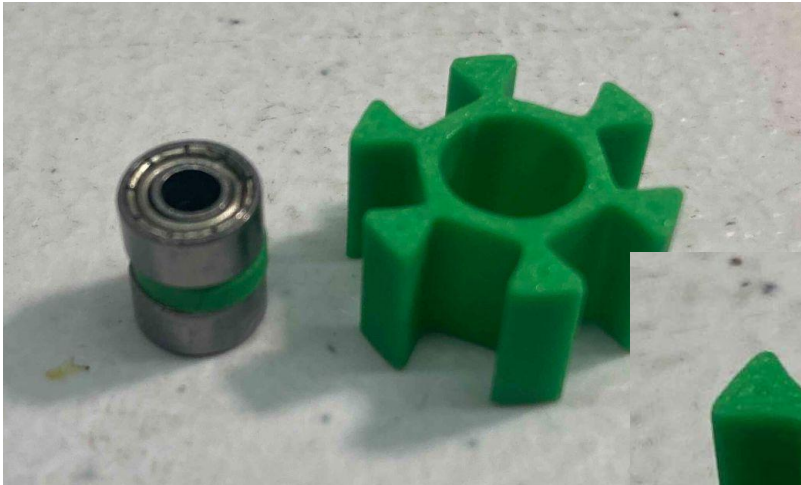


**put bearing+ring+¹
bearing sandwich
into the magnet
holder**

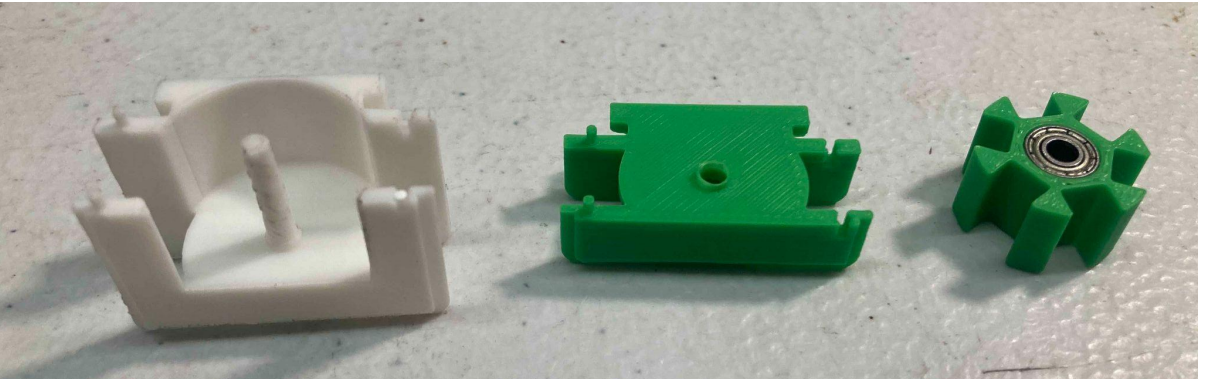
**one at a time!
no magnets in here for a
while... :-)**



magnet ring assembly

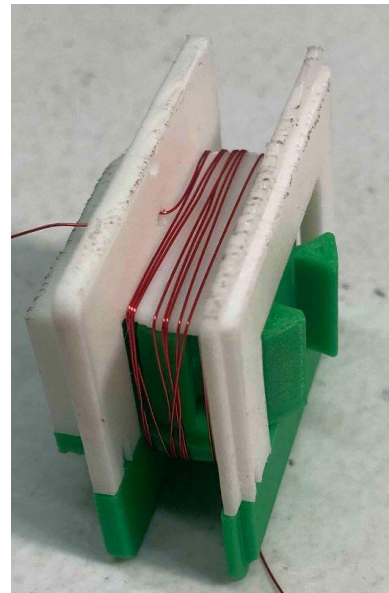
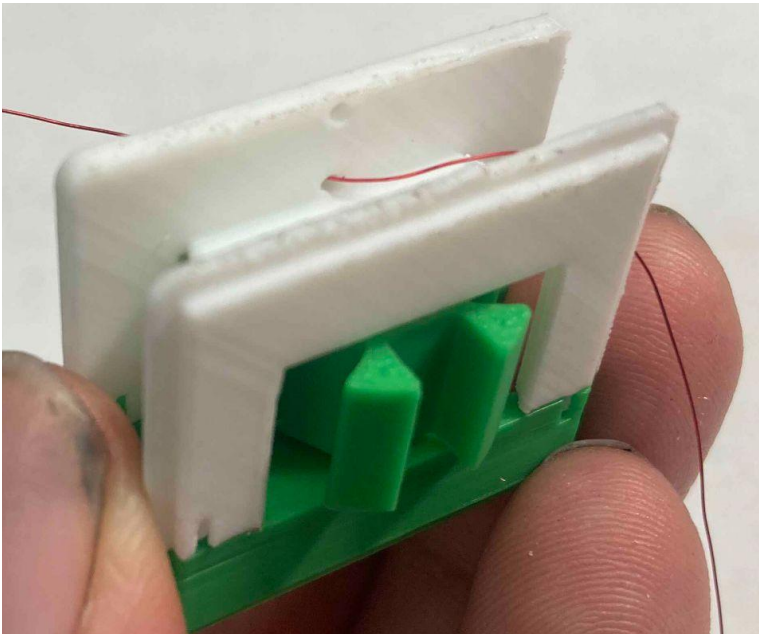
2

- put magnet ring onto bottom bobbin shaft
- then add top bobbin

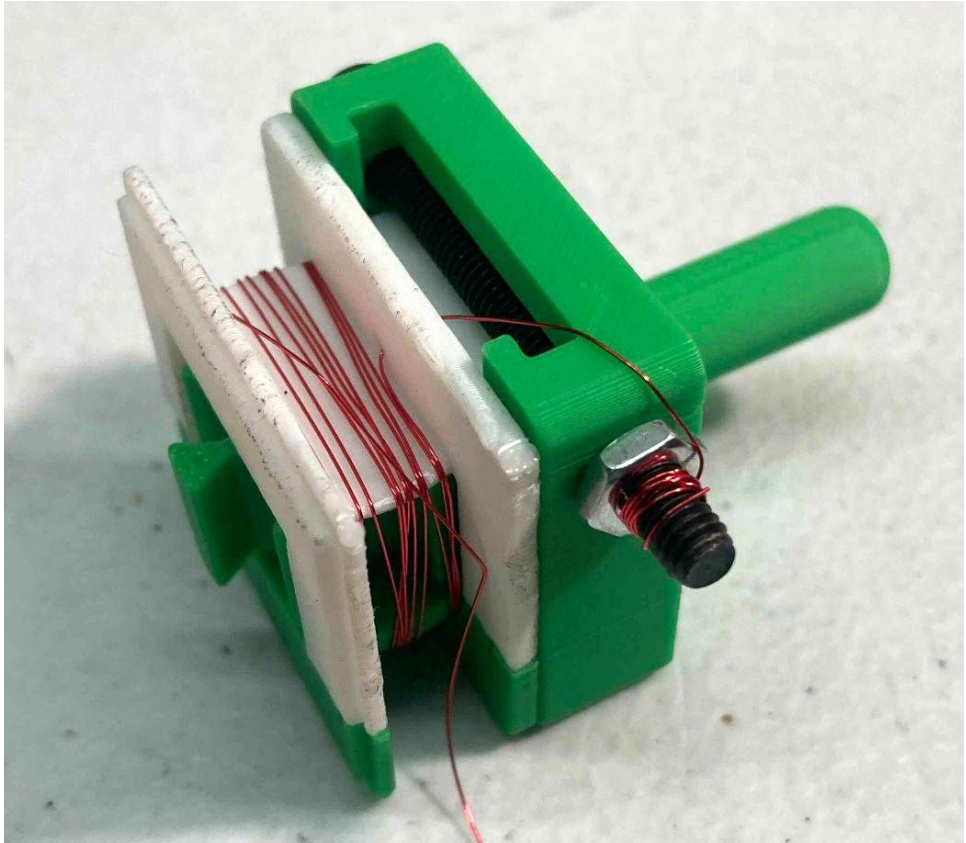


Wire Prep

- feed magnet wire through **lower hole** in bobbin (notice the hole above it too!)
- pull ~5-6" through
- do about 20 turns by hand. make the loops tight, not loose!



- **slide this assembly into the winding attachment.**
- **put in the M4 screw and nut (hand tight) to lock.**
- **wrap the excess 6" of wire around the end of the screw so it is not in the way**



Coil Pre-Wind!

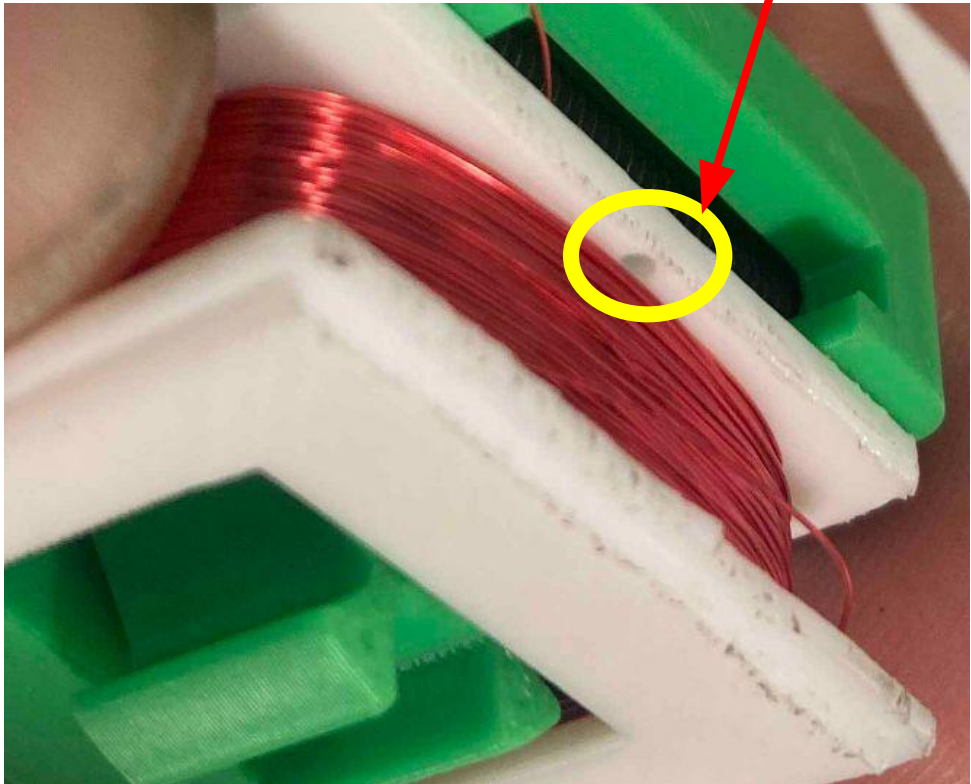
5

- load the previous assembly into a drill chuck
- have a friend hold magnet wire spool, or put on the spool holder
- we need to add tension to the wire and guide it onto the bobbin evenly,
- I typically use my hand for this, so you usually need to **hold the wire between your fingers** and add tension and guide it left to right. (you can also use a piece of plastic or paper btw your fingers and the wire)



Coil Winding!

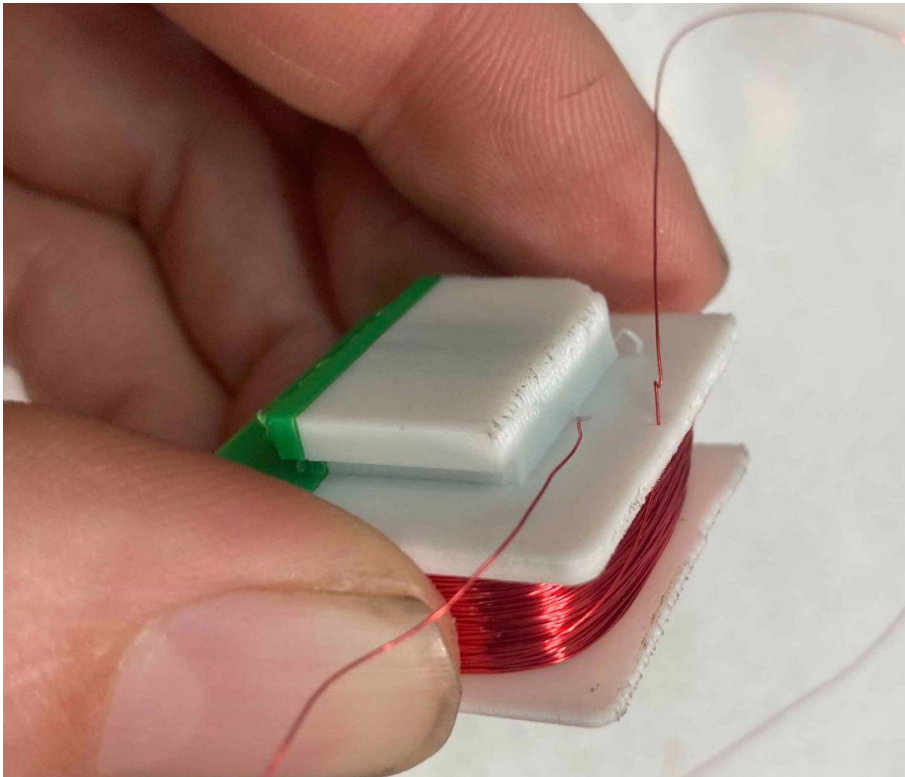
- once you have everything ready, start spinning the bobbin. make sure you are spinning in the right direction! **go verrrrry slow at first!**
- The goal is tight windings, evenly distributed across the bobbin.
- Guide the wire left to right to left slowly as the bobbin rotates
- Wind until you are **just below the upper hole** in the bobbin. (above where you fed in the initial wire).
- stop and cut the wire about 8" long



remove bobbin from attachment

7

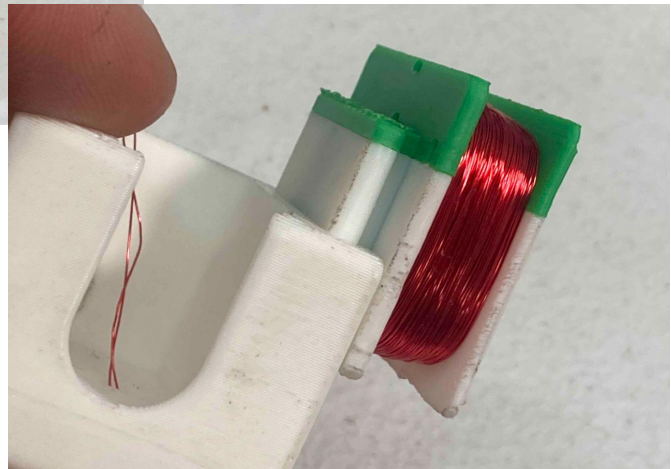
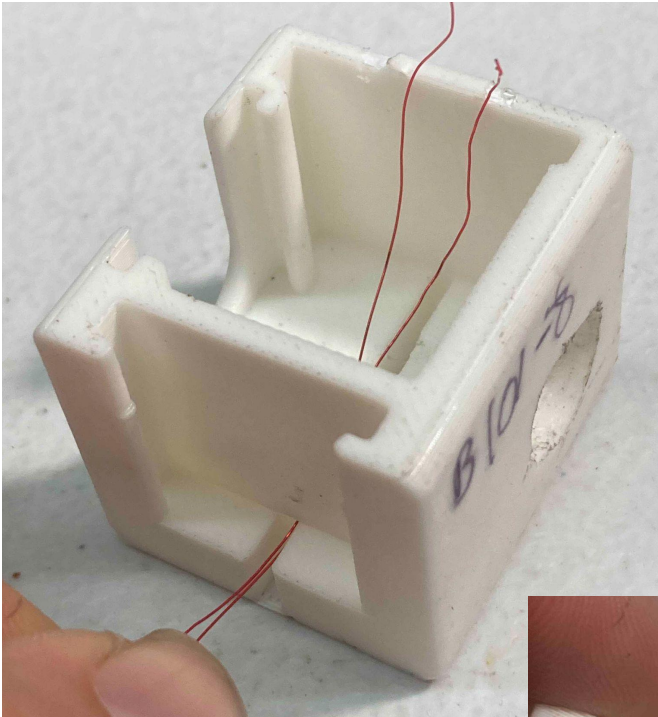
- **unwind excess wire from screw**
- **take out nut, screw**
- **remove bobbin.**
- **wind the wire around a couple times tightly.**
- **feed it through the upper hole**



Coil Installation

8

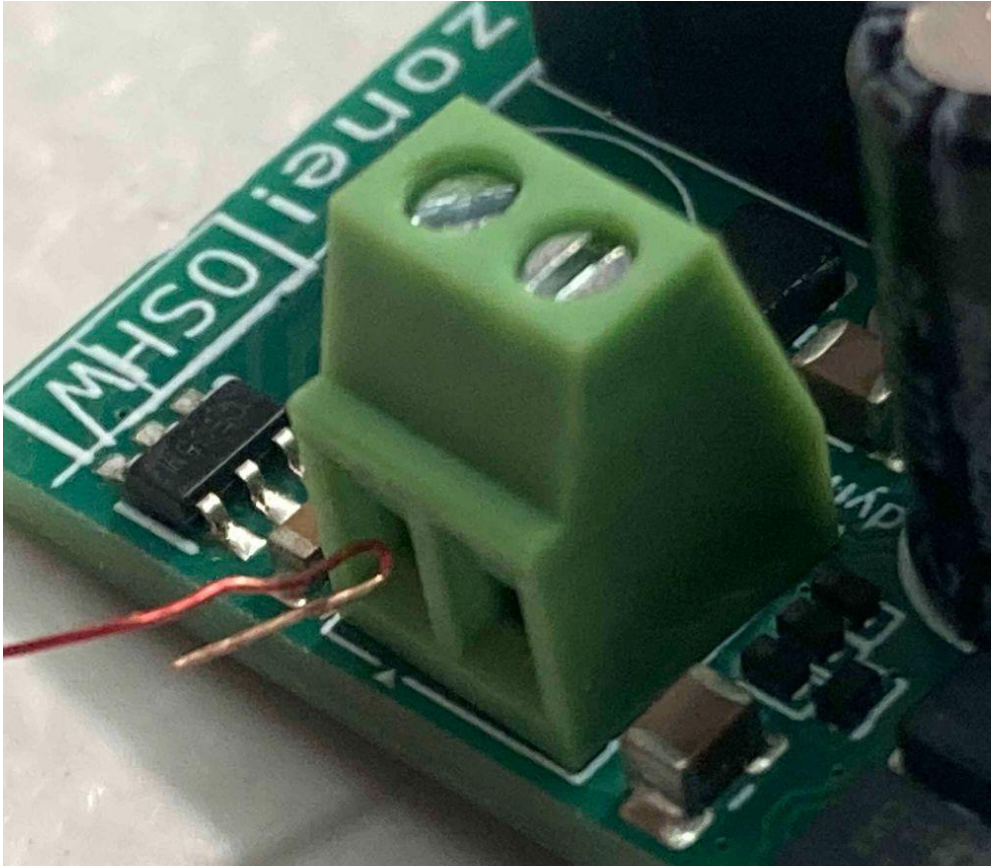
- cut wire down to ~ 3".
- feed wire through the slot in bottom of the main enclosure
- slide bobbin onto enclosure with wires coming out bottom



COIL PCB wiring

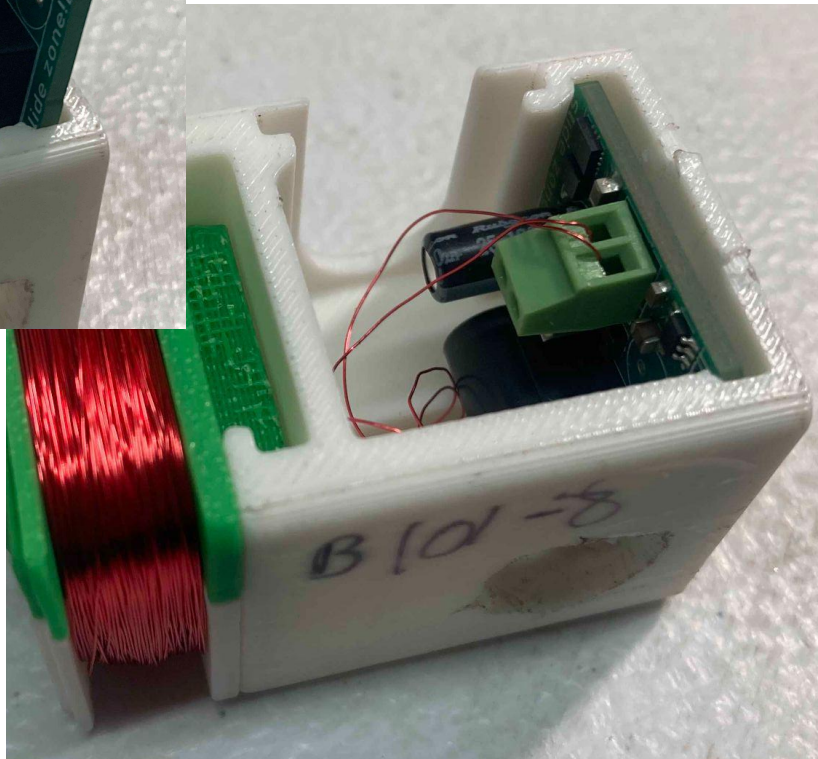
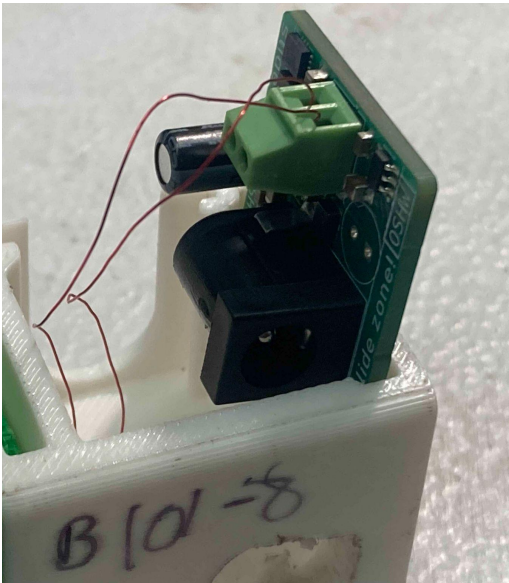
9

- use sandpaper to sand off about $\frac{1}{4}$ " off the ends of the wires.
- the color should wipe off and you see bare copper.
- Loosen the small screw (using the small flathead screwdriver) on the green connector on the rectifier PCB.
- We will be putting the wires into this connector.
- fold about $\frac{1}{8}$ " back of each wire. put the wire into one of the small holes on the side of the green connector. tighten.
- Add the other wire to the remaining hole on the connector.



PCB Installation

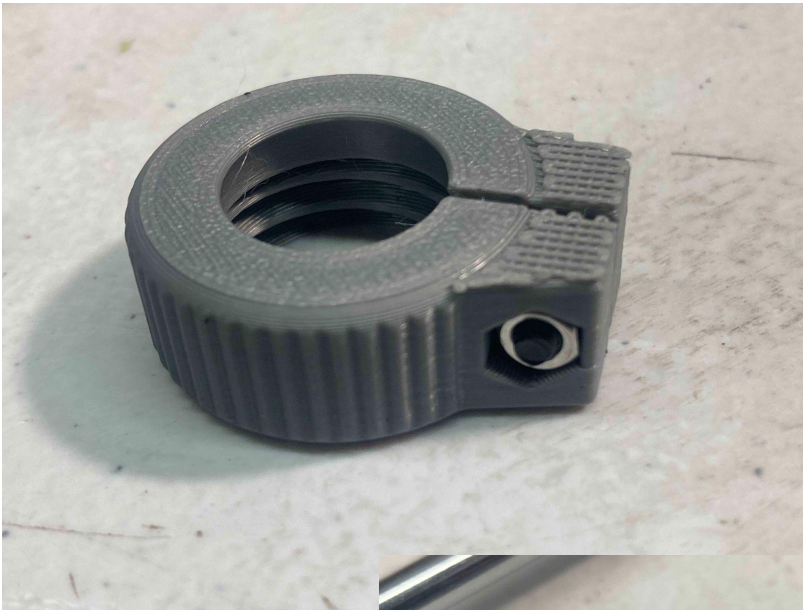
- Side PCB into the rear of the enclosure, making sure the green connector is facing up!



Assemble dynamo mount!

11

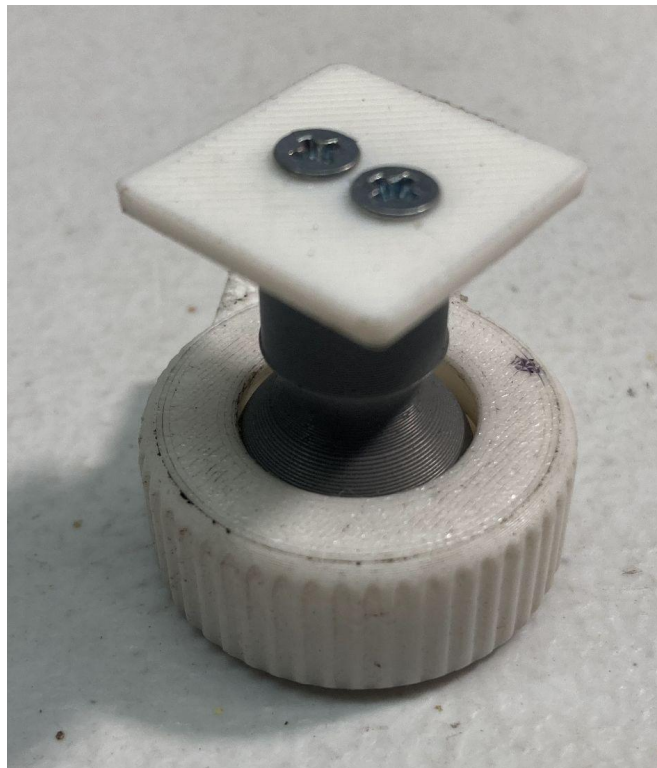
- Add M3x16 screw and M3 nut on the large 3D printed nut.
- Put the 3D printed nut around the shaft
- Make sure it it in can go around the ball nicely.
- If it's upside down, you can see the threads.



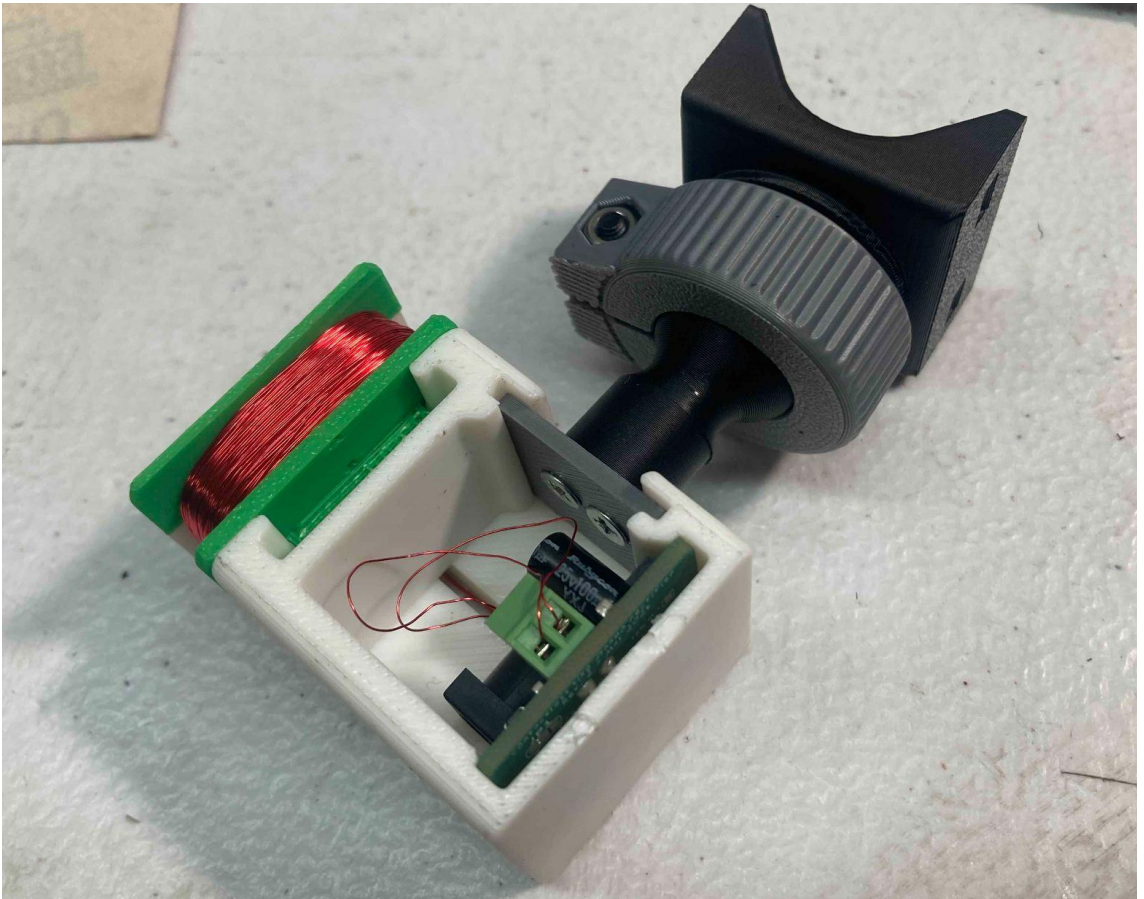
Assemble dynamo mount!

12

- Add the plate and 2 screws.
- Make sure the screws go through the side of the plate that is countersunk.



- **Screw in the ball nut into the bike mount.**
- **slide this assembly onto the enclosure**



Add Magnets to Magnet Ring! 14

- **caution!** These are very strong!
- 2 magnets per slot.
- make sure each pair of magnets is opposite polarity from the one before.
- test by holding the pair close to the previous one and seeing if they attract or repel.
- if they repel that is good, put them into the next slot that way. if not, flip them.



Assemble Light!

15

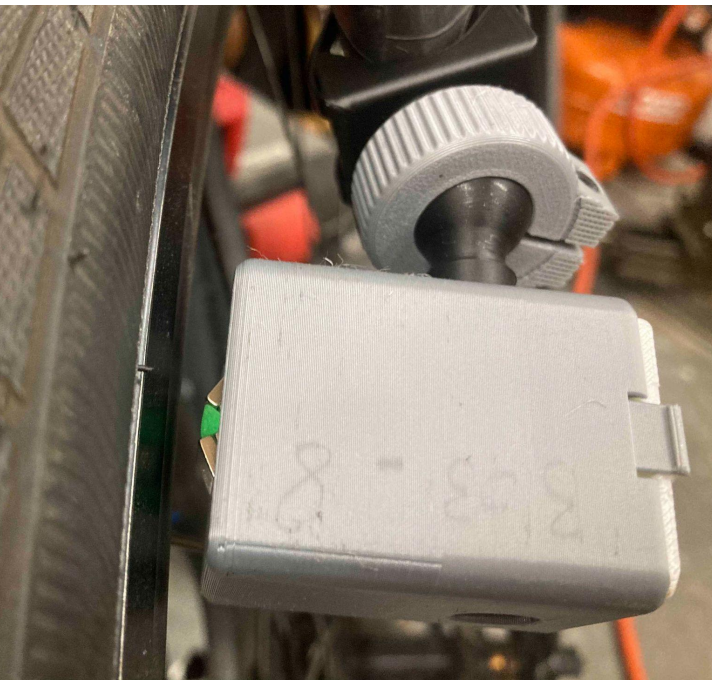
- Add Light PCB to the enclosure.
- Stick the lens into the small holes on the lid.
- Slide the lid onto the case.
- There is a way to get red light out of this PCB, but it is annoying, so we will just be using white light. If you really want red, it requires some delicate knife work and a few wires to solder. maybe wait for V2....



Mounting!

16

- Find a good place for the dynamo to live on a fork.
- The magnets need to be about 3mm - 5mm away from the rim.
- Add some bike inner tube to that area and tape it down.
- Then mount the dynamo with zip ties
- Mount light somewhere on the bike with zip ties!
- Connect the two with the 3' cable.
- Secure cable with zip ties.
- **LIGHT UP THE NIGHT WITHOUT SOME FREE ENERGY! AND NO DANG BATTERIES !**



contactless bicycle ¹⁷ dynamo workshop!

Make a bicycle light
without a battery!

